

SMART: Shot-Aware Multimodal Video Moment Retrieval with Audio-Enhanced MLLM

An Yu, Weiheng Lu, Jian Li, Zhenfei Zhang, Yunhang Shen, Felix X.-F. Ye, Ming-Ching Chang

Abstract—Video Moment Retrieval aims to localize a target temporal segment in an untrimmed video given a natural-language query. Despite recent progress in both traditional approaches and multimodal large language model (MLLM)-based methods, many existing methods still rely primarily on visual cues and only weakly exploit temporal structure, which limits performance on complex videos. To address this issue, we introduce Shot-aware Multimodal Audio-enhanced Retrieval of Temporal Segments (SMART), an MLLM-based framework that jointly leverages audio cues and shot-level temporal structure. SMART enriches multimodal representations by incorporating complementary audio information, and introduces a Shot-aware Token Compression strategy that preserves representative keyframes as full frames while compressing only non-keyframes to reduce visual redundancy. We also refine the prompting strategy to better utilize audio-visual context. Experiments on Charades-STA and QVHighlights show that SMART consistently improves over strong baselines and achieves favorable efficiency-effectiveness trade-offs, including gains of 1.61% in R1@0.5 and 2.59% in R1@0.7 on Charades-STA.

Index Terms—Video Moment Retrieval, Temporal Localization, Audio-Visual Representation Learning, Shot-aware Token Compression, Shot Boundary Detection, Temporal Reasoning, Multimodal Large Language Models (MLLM), Video Understanding.

I. INTRODUCTION

WITH the rapid growth of video content shared and created on the internet and social media, the ability to efficiently analyze such content has become increasingly important. One key task in this domain is **moment retrieval**—the process of identifying the specific temporal segment within a video that best corresponds to a given natural language query. This task is fundamental to a wide range of applications, including video summarization, content recommendation, security surveillance, and event localization. It has emerged as a prominent area of research due to its practical importance and inherent complexity.

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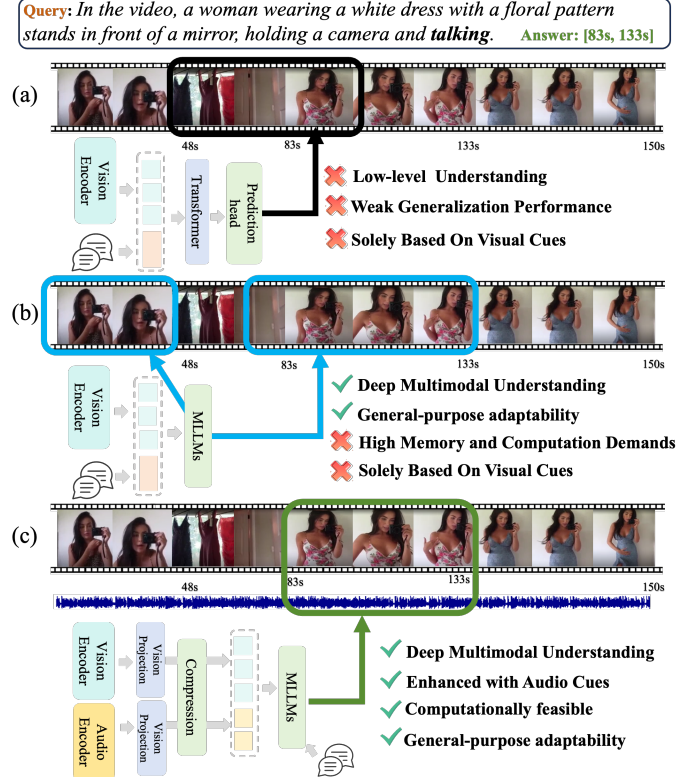


Fig. 1. **Overview:** (a) Traditional models rely on low-level visual features and generalize poorly. (b) MLLM-based models improve semantic understanding but ignore audio and incur high computational cost. (c) Our SMART model integrates audio and shot-coherent token compression for efficient, accurate multimodal moment retrieval.

Current moment retrieval methods can be broadly categorized into two main classes: traditional approaches [3], [8], [7], [56], [57] and approaches that leverage Multimodal Large Language Models (MLLM) [45], [46]. As shown in Figure 1(a), traditional approaches typically adopt Transformer-based architectures with task-specific prediction heads. These methods often rely on low-level visual features and are tailored to specific tasks, which limits their ability to generalize and capture high-level semantics. In contrast, recent studies [47], [14], [17], [9], [55] have explored the use of MLLM to enhance semantic understanding for moment retrieval tasks. However, as illustrated in Figure 1(b), these methods typically rely solely on visual information and overlook the audio modality, which can provide complementary evidence and is particularly beneficial for queries involving sound or temporal continuity. For example, retrieving the moment for the query “a woman is holding a camera and talking” requires not only visual identification but also speech detection. Similarly,

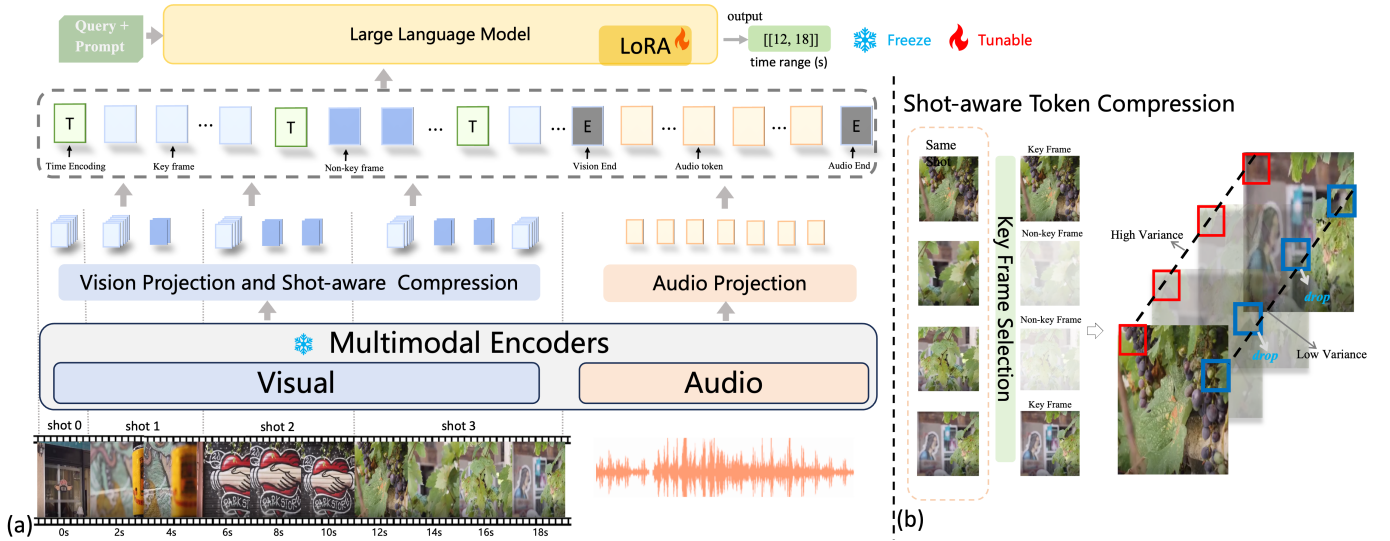


Fig. 2. The **Shot-aware Multimodal Audio-enhanced Retrieval of Temporal Segments (SMART)** architecture: (a) The SMART pipeline integrates a pretrained MLLM with frozen visual and audio encoders and a lightweight LoRA-tuned LLM to predict temporal segments relevant to a query. Visual and audio features are projected, temporally encoded, and concatenated into a unified multimodal prompt, with shot-aware token compression applied for efficiency. (b) The Shot-aware Token Compression (STC, detailed in Fig. 3) operates in two stages: frames are first classified into key and non-key frames based on inter-frame differences; then, token-wise variance within each shot is analyzed. High-variance tokens (red), representing dynamic content, are retained, while low-variance tokens (blue) from non-key frames are discarded to remove redundancy while preserving essential cues.

the sound of a siren is often key to identifying moments involving passing vehicles. Without audio, models may miss complementary cues that are helpful for temporally ambiguous queries. This unimodal assumption poses challenges, especially when temporal grounding depends on acoustic cues or joint audio-visual signals.

LLMs often face challenges when handling long input sequences, which restricts the amount of video content they can effectively process and leads to high memory and computational demands. As a result, they may fail to capture the complete and nuanced temporal context spread across extended video sequences, which is essential for accurately identifying and retrieving the segments that align closely with a given natural language query. The inefficiency in handling long sequences also poses a major scalability challenge, especially in real-world applications where videos are lengthy and complex. These constraints reduce their capacity to capture rich, multimodal temporal context, thus limiting their effectiveness in tasks that require precise and robust video understanding and moment retrieval.

To address limitations in multimodal modeling and efficiency for long-video moment retrieval, we propose **Shot-aware Multimodal Audio-enhanced Retrieval of Temporal Segments (SMART)**—a novel MLLM-based framework illustrated in Figure 1(c)). SMART integrates audio information and employs a dual-branch encoder to jointly process video and audio inputs, enriching multimodal representations and enabling more comprehensive temporal understanding. To reduce the computational cost of long input sequences, we introduce **Shot-aware Token Compression (STC)** that compresses redundant frame tokens within each shot. By computing token-level variance across frames using Q-Former outputs, it identifies and retains key tokens while discarding redundant ones. This compression aims to preserve semantic

richness and fine-grained temporal cues, allowing SMART to process longer video sequences more efficiently while reducing redundant visual tokens. The complete pipeline is shown in Figure 2.

We conduct extensive experiments on two widely used benchmark datasets, Charades-STA [1] and QVHighlights [3]. The results show that SMART consistently improves over strong baselines by better leveraging complementary audio cues and reducing visual redundancy through shot-aware compression. On the QVHighlights test set, SMART improves R1@0.5 by 1.56% over LLaVA-MR. On Charades-STA, it improves R1@0.5 by 1.61 points and R1@0.7 by 2.59% over LLaVA-MR.

The main contributions of this paper are summarized as follows:

- We propose SMART, a multimodal large language model (MLLM)-based framework for moment retrieval that incorporates complementary audio cues to support temporal localization.
- We introduce a shot-aware token compression strategy that preserves representative keyframes as full frames and compresses only non-keyframes, thereby reducing visual redundancy while retaining informative temporal evidence.
- We conduct comprehensive experiments on Charades-STA and QVHighlights, showing that SMART achieves consistent performance gains and favorable efficiency–effectiveness trade-offs over strong baselines.

II. RELATED WORK

A. Video Moment Retrieval

Video Moment Retrieval (VMR), also known as temporal sentence grounding in videos, aims to localize the video

segment that best matches a natural-language query. Early methods such as TALL [1] formulated this task as sliding-window matching, while attention-based designs improved efficiency by directly predicting temporal boundaries [19]. Subsequent approaches strengthened video-text alignment through query-conditioned modulation and structured reasoning, as represented by Semantic Conditioned Dynamic Modulation [20] and the Moment Alignment Network [21]. More recent studies have explored stronger temporal modeling, supervision-efficient localization, and generalization-oriented settings, including frame-supervised localization [58] and zero-shot VMR [59]. Despite these advances, most VMR methods still focus primarily on visual-text matching and treat videos as frame sequences or clip sequences without explicitly modeling shot-level organization. This can introduce substantial visual redundancy, especially for long videos with repeated or visually similar frames. SMART addresses this gap by integrating audio cues and shot-aware compression within an MLLM-based retrieval framework, improving temporal grounding while reducing unnecessary visual tokens.

B. Audio-Enhanced Video Understanding and Retrieval

Audio provides complementary temporal and semantic evidence for video understanding, especially when the target event is only partially visible or when key cues are conveyed by sound. Recent audio-aware VMR work [60] shows that acoustic information can materially improve localization for sound-related queries. Beyond VMR, audio-visual event perception and parsing methods further demonstrate the value of acoustic cues in noisy or weakly supervised settings [61], [62]. However, audio is not uniformly beneficial: its utility depends on whether the query is acoustically grounded, whether the sound is temporally aligned with the visual event, and how audio is fused with visual and textual information. Different from prior audio-enhanced methods, SMART studies audio within an MLLM-based retrieval pipeline and combines it with shot-aware visual compression. This design allows the model to retain richer multimodal context while limiting redundancy in long video sequences.

C. Robustness and Debiasing in Temporal Grounding

Robustness and debiasing are also closely related to temporal grounding. Existing datasets and models may contain temporal shortcuts, annotation artifacts, or evaluation biases, causing models to exploit spurious correlations rather than genuine temporal reasoning. Prior studies have therefore examined dataset bias, evaluation bias, counterfactual analysis, and debiasing strategies for temporal sentence grounding [63], [64], [65], [66]. These works show that strong in-distribution performance does not necessarily indicate robust temporal understanding.

Although SMART does not directly target dataset debiasing, its design is related to robustness in two ways. First, shot-aware compression reduces dense and redundant visual evidence, encouraging the model to focus on temporally informative content instead of overusing frame-level repetition. Second, our analyses examine sensitivity to shot-boundary

perturbations and coarse audio temporal shifts, which provides insight into how multimodal temporal grounding behaves under structured perturbations and modality-specific noise. This is important because small changes in temporal segmentation or modality alignment may alter the evidence available to the model without changing the underlying query intent. Recent robust and data-efficient adaptation studies further show that noisy samples, channel bias, and feature redundancy can reduce the reliability of pretrained visual representations [68], [69]. Although these works mainly study few-shot visual recognition rather than VMR, they support our motivation to reduce noisy or redundant evidence through structured compression.

D. Multimodal LLMs for Video Understanding

Building on the success of large language models, Multimodal Large Language Models (MLLMs) have become a powerful paradigm for cross-modal reasoning. Existing MLLMs extend LLMs with gated cross-attention, adapter modules, projection layers, or Q-Formers to bridge visual and textual representations [29], [30], [31], [32], [33], [34], [35]. Video-oriented MLLMs further adapt these mechanisms to temporal inputs and support video-level reasoning [36], [39], [38], [37], [51]. Recent retrieval-oriented models such as Mr. BLIP [9] and LLaVA-MR [17] demonstrate that MLLMs can improve semantic alignment for moment retrieval. In parallel, prompt-tuning methods study parameter-efficient adaptation and base-to-new generalization for vision-language models [70], which is complementary to our focus on audio-enhanced VMR with shot-aware compression. Nevertheless, directly applying MLLMs to long videos remains challenging because dense visual tokens increase computation and may dilute temporally relevant evidence.

Our design is also related to ActBERT [67], which learns joint video-text representations by combining global action information with local object-level regions. Although ActBERT predates recent MLLMs and does not address audio-enhanced moment retrieval, its global-local design philosophy is aligned with SMART’s goal of preserving coarse temporal organization while reducing local redundancy. Unlike prior MLLM-based retrieval methods that mainly emphasize semantic alignment, SMART further incorporates explicit audio integration and shot-aware visual compression. As a result, it seeks a better balance between multimodal reasoning, temporal structure preservation, and computational efficiency for video moment retrieval.

III. METHOD

We advance MLLMs for moment retrieval by incorporating audio as complementary contextual information alongside visual inputs, together with a token compression mechanism to reduce sequence length. The following sections present the overall SMART architecture (§ III-A), describe the audio integration strategy that enhances multimodal understanding (§ III-B), and introduce a shot-aware token compression approach designed to reduce computational cost while preserving fine-grained temporal detail (§ III-C).

A. Pipeline Design

Our model architecture is illustrated in Fig. 2. The model uses two separate branches for visual and audio feature extraction. In the visual branch, EVA-CLIP [27] is adopted as the visual encoder to process sampled frames and produce a sequence of image embeddings, denoted as $\mathbf{V} = [\mathbf{v}_1, \dots, \mathbf{v}_N] \in \mathbf{R}^{N \times P \times D_v}$, where N is the number of frames, P is the number of patches per frame, and D_v is the visual feature dimension. A query-based transformer (Q-Former) then refines each frame embedding to generate frame-level features $\mathbf{V}^q = [\mathbf{v}_1^q, \mathbf{v}_2^q, \dots, \mathbf{v}_N^q] \in \mathbf{R}^{N \times Q \times D_q}$, with Q queries and a Q-Former output dimension of D_q . The output, $\mathbf{V}^q$, is projected into the LLM feature space via a vision projection layer. The shot-coherent token compression module then reduces token redundancy, producing the final visual representation $\mathbf{V}^L \in \mathbf{R}^{S_v \times D_L}$, where S_v is the number of compressed visual tokens and D_L is the LLM input dimension.

In the audio branch, we adopt BEATs [28], an acoustic tokenizer that produces discrete embeddings rich in semantic information. The audio embeddings are denoted as $\mathbf{A} = [\mathbf{a}_1, \dots, \mathbf{a}_T] \in \mathbf{R}^{T \times D_a}$, where T is the number of audio tokens and D_a is the audio feature dimension. Given the continuous but often redundant nature of audio data, we apply average pooling to reduce the sequence length, followed by an audio projection layer that maps the audio features into the LLM space. The final audio representation is $\mathbf{A}^L \in \mathbf{R}^{S_a \times D_L}$, where S_a is the pooled length of audio tokens.

To construct the input for the language model, we organize a multimodal sequence consisting of time tokens t_i , visual tokens $\mathbf{v}_i^L \in \mathbf{V}^L$, audio tokens $\mathbf{a}_i^L \in \mathbf{A}^L$, a task-specific query q , and a prompt p . To provide explicit temporal context, each time token is placed directly before its corresponding frame embedding. We assign explicit temporal tokens only to visual frame embeddings, while audio features are incorporated as pooled acoustic context rather than being explicitly aligned with individual visual timestamps. Additionally, to distinguish between modalities, we insert special separator tokens: V_E to mark the end of visual embeddings and A_E for the end of audio embeddings. The resulting multimodal input sequence is structured as:

$$x = [t_1, \mathbf{v}_1^L, t_2, \mathbf{v}_2^L, \dots, V_E, \mathbf{a}_1^L, \mathbf{a}_2^L, \dots, A_E, q, p] \quad (1)$$

This token sequence is then fed into the LLM to identify time segments relevant to the input query q . Inspired by the design of Mr. BLIP [9], the model is prompted to generate a sequence of one or more temporal moments, each represented by a start and end timestamp in seconds. The output follows a nested list format, e.g., $[[s_1, e_1], [s_2, e_2], \dots]$.

B. Audio Integration

Current mainstream benchmark datasets for moment retrieval predominantly provide textual descriptions grounded in visual content, leading most existing methods to rely primarily on visual cues. However, audio can offer complementary information to the visual scene; for instance, footsteps may provide useful acoustic evidence for walking-related actions. Incorporating such acoustic cues enables richer multimodal

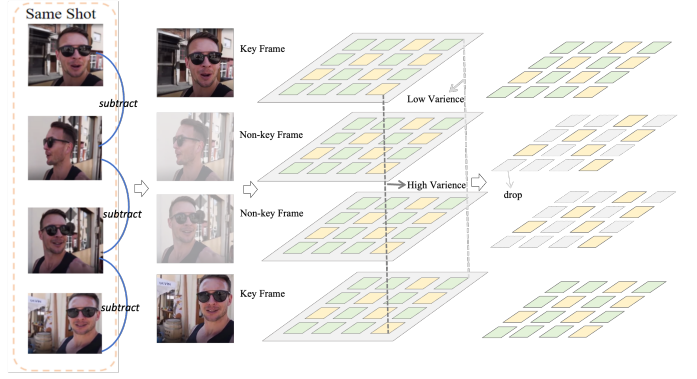


Fig. 3. **Shot-aware token compression:** Key frames are detected via inter-frame differences. Within each shot, high-variance tokens from dynamic regions are preserved, while low-variance tokens from non-key frames are discarded, effectively minimizing redundancy and retaining essential temporal cues for accurate moment retrieval.

reasoning and can improve temporal localization, particularly in cases where visual evidence alone is insufficient or temporally ambiguous.

To incorporate audio, we introduce an efficient module that extracts semantic acoustic cues and injects them as complementary context into the vision-language framework. Raw audio is extracted and resampled to a standardized 16kHz rate, then encoded using the pretrained BEATs [28] model, which is known for its strong semantic understanding of audio signals. To reduce redundancy and maintain computational efficiency, the audio features are temporally pooled into a compact representation and projected to match the visual feature dimension. This design facilitates seamless cross-modal integration, allowing the model to leverage auditory cues (such as speech or environmental sounds) for improved temporal grounding.

Notably, we do not explicitly insert timestamp tokens into the audio features. Instead, audio is incorporated as complementary contextual information rather than an explicitly time-aligned stream. BEATs provides temporally ordered acoustic representations through frame-wise encoding and positional embeddings, which are sufficient for our design without requiring explicit fine-grained audio-visual alignment. We further explore and compare different audio integration strategies in the Ablation Study in § IV-C.

C. Shot-aware Token Compression (STC)

We propose the **Shot-aware Token Compression (STC)** strategy to efficiently process long multimodal videos while preserving fine-grained temporal cues. The key insight is that frames within the same shot exhibit strong visual similarity and temporal coherence, introducing redundancy that can be exploited for compression. STC addresses this by (1) identifying keyframes that capture meaningful scene dynamics, and (2) performing intra-shot token compression that removes redundant visual tokens from non-keyframes. This design substantially reduces sequence length while maintaining essential spatio-temporal semantics for accurate moment retrieval. For guiding the compression process, we first segment each video into distinct shots using TransNetV2 [18], which detects shot boundaries based on learned visual transition cues.

Motivation: Dense frame sampling is crucial for capturing subtle visual transitions and transient events, yet it greatly increases sequence length—particularly when combined with audio inputs. Transformer-based LLMs suffer from quadratic self-attention complexity, resulting in high memory cost and degraded inference efficiency for long sequences. Thus, effective token compression is necessary to remove redundancy while preserving discriminative temporal and semantic information. Existing approaches, such as clustering or pooling [48], [49], often lose fine details in dynamic scenes. In contrast, STC performs compression at the shot level, where intra-shot visual redundancy is high and temporal coherence is preserved, ensuring better efficiency–accuracy trade-offs.

Stage 1 Keyframe Identification: Given visual features $\mathbf{F}^v$ extracted from the image encoder, we compute frame-to-frame differences to detect visually dynamic frames: $\Delta \mathbf{V} = [\Delta \mathbf{v}_1, \dots, \Delta \mathbf{v}_N]$, where each $\Delta \mathbf{v}_i$ encodes patch-wise changes between consecutive frames. The L2 norm of each $\Delta \mathbf{v}_i$ yields a motion magnitude sequence $\mathbf{d} = [d_1, d_2, \dots, d_N]$. To suppress local jitter and noise, we apply a Gaussian filter to obtain a smoothed series $\hat{\mathbf{d}} = \text{Gaussian}(\mathbf{d})$, where each $\hat{d}_i$ represents the smoothed motion magnitude of the i -th frame. Frames with higher $\hat{d}_i$ values are selected as keyframes that contain dynamic content, while the remaining frames are considered non-keyframes with redundant information suitable for compression.

Stage 2 Intra-Shot Token Compression: For each shot, token-level compression is performed within the Q-Former structure, which uses Q learnable queries to encode each frame. Each query captures distinct semantic aspects of the visual content. For all frames in a shot, token variance across the temporal dimension is computed at each query position. High-variance tokens correspond to dynamic elements and are retained across all frames. Low-variance tokens, typically representing static background or redundant content, are discarded from non-keyframes. Their information can be inferred from corresponding tokens in keyframes. This selective compression reduces visual redundancy and computational load, while preserving the temporal fidelity necessary for precise multimodal moment retrieval. Figure 3 illustrates the STC process.

IV. EXPERIMENTAL EVALUATION

A. Experiment Setting

Benchmarks: We validate our model on two widely-used video Moment Retrieval (MR) datasets: (1) **Charades-STA** [1] contains 9,848 videos with an average duration of 30.6 seconds. It provides 16,128 annotations (12,408 for training and 3,720 for testing), where moments average 8.1 seconds in length and queries average 7.22 words. (2) **QVHighlights** [3] consists of 10,148 YouTube videos, each 150 seconds long. Every video includes at least one annotated query (averaging 11.3 words) describing relevant moments, which average 24.6 seconds in duration. The dataset is split into 7,218 training queries and 1,150 validation queries.

Although ActivityNet Captions [44] is widely used for moment retrieval, we do not include it in our experiments because the commonly used official release

(“Anet_videos_15fps_short256”) contains silent videos only, which prevents us from evaluating the audio branch of SMART in its intended multimodal setting. Similarly, datasets such as TACoS [52] and MAD [53] are typically provided as pre-extracted visual features without access to raw video frames or audio streams. Since SMART relies on frame-level visual processing, shot detection, and audio encoding as part of the full pipeline, these datasets do not support a fair and reproducible evaluation of the complete method. We therefore focus our experiments on benchmarks that allow the full multimodal pipeline to be executed. In particular, QVHighlights serves as our main long-video benchmark for analyzing the proposed audio design and shot-aware token compression.

Evaluation Metrics: We evaluate Moment Retrieval (MR) using standard metrics: Recall@K, mean Average Precision (mAP), and mean Intersection over Union (mIoU), computed at specific IoU thresholds. Recall@K measures the proportion of top-K predictions with IoU above a threshold (e.g., R1@0.5). mIoU calculates the average IoU across all predictions, while mAP measures the mean precision across multiple IoU thresholds (e.g., mAP@0.5, mAP@0.75). In summary, R1@0.5 and R1@0.7 assess retrieval accuracy at different IoU levels, mIoU captures average segment overlap, and mAP reflects overall localization precision across thresholds.

Implementation Details: We adopt parameter-efficient fine-tuning with LoRA [43], updating only 0.63% of the model parameters. To mitigate occasional inconsistencies in LLM outputs, we apply lightweight post-processing heuristics to correct minor formatting errors and improve prediction reliability. Training is performed with the AdamW optimizer [2], starting from a learning rate (LR) of $1e-8$, linearly warmed up to $3e-4$ over the first 10% of iterations, and subsequently decayed with a cosine schedule. During training, frames are randomly sampled from each video.

For Charades-STA, we sample 60 frames per video and train for 30 epochs with a batch size of 16 using 4 GPUs. For QVHighlights, we sample 80 frames per video and train for 30 epochs with a batch size of 32 on 8 GPUs, applying gradient accumulation with a factor of 4. All experiments are conducted on NVIDIA A100 GPUs with 80GB memory.

B. Results

Comparison to the State of the Art: We evaluate our approach on two widely used moment retrieval datasets: *Charades-STA* and *QVHighlights*. As shown in Table I, our model consistently surpasses prior methods across nearly all evaluation metrics on both benchmarks, demonstrating strong generalization regardless of dataset complexity or evaluation scenario.

Our proposed model, **SMART**, achieves substantial performance gains on both datasets, highlighting its robustness across videos ranging in duration from a few seconds to several minutes.

QVHighlights: On the validation set, SMART surpasses previous approaches with improvements of **0.52%** in R1@0.5 and **0.9%** in R1@0.7. On the test set (Table I), SMART achieves the best results among the compared methods across the major reported metrics, achieving an R1@0.5 of **78.15**, R1@0.7 of



Fig. 4. **Qualitative results on QVHighlights [3]:** This figure visualizes the predicted and ground-truth segments for query events. SMART outperforms the visual-text MLLM baseline by preserving shot-level consistency and leveraging audio cues for fine-grained temporal understanding. See text for details.

63.16, mAP@0.5 of **70.76**, and mAP@0.75 of **55.54**. These results represent consistent and notable improvements over the strongest existing baselines, including LLaVA-MR and Mr.BLIP. On the validation set, SMART also achieves the highest mIoU of **72.03**, outperforming all competing architectures built on BLIP and CLIP backbones. This demonstrates SMART’s superior ability to model fine-grained temporal structure and multimodal semantics.

Charades-STA: On the Charades-STA test set, SMART also achieves clear performance gains, improving R1@0.5 by **1.61%** and R1@0.7 by **2.59%**. It further attains a new best mIoU of **61.09**, outperforming the previous leading method, Mr.BLIP (58.63), as well as other strong CLIP-based competitors such as CG-DETR and VideoLights-B-pt. These consistent improvements across diverse benchmarks validate the effectiveness of our design choices, particularly the integration of LLM-driven semantic alignment with multimodal supervision. Overall, SMART achieves strong performance over prior methods across both short- and long-duration video benchmarks.

Qualitative Results: Fig. 4 illustrates qualitative comparisons between ground-truth query segments and predicted intervals

from baseline MLLM-based models (without audio) and our proposed SMART. In Example 1, competing MLLM models fail to maintain temporal precision due to the absence of shot-coherent token compression. In contrast, SMART leverages shot boundaries to retain the most relevant tokens, preserving contextual consistency within each shot and mitigating interference across shots. Moreover, by incorporating audio cues such as character dialogues, SMART captures fine-grained semantics more effectively, for example in the case of “*sharing wired earplug headphones*”. These advantages enable SMART to deliver substantially more accurate predictions than competing approaches.

In Example 2, other MLLM-based models incorrectly predict the interval 78s–82s and miss the correct moment around 92s. This misalignment likely results from losing fine-grained visual cues—such as the presence of “*two men*” during feature compression. In contrast, SMART applies shot-coherent compression that preserves essential information within each shot while filtering redundancy, enabling it to correctly identify the 92s–106s interval, which closely aligns with the ground truth.

In Example 3, for the query “*A man talking from a beach*”, the keyword talking clearly requires audio understanding.

TABLE I
COMPARISON WITH STATE-OF-THE-ART METHODS ON THE
CHARADES-STA AND QVHighlights DATASETS. **CLIP***: UNLOC [4]
PRETRAINED ON KINETICS 400/700 ACTION CLASSIFICATION, **SF**:
SLOW-FAST BACKBONE.

Method	Backbone	mIoUR1@0.5R1@0.7mAP@0.5mAP@0.75
QVHighlights Validation set		
Moment-DETR [3]	SF+CLIP	59.68 40.84 – –
EaTR [5]	I3D	61.36 45.79 61.86 41.91
QD-DETR [6]	SF+CLIP	62.68 46.66 62.23 41.82
UVCOM [11]	SF+CLIP	65.10 51.81 – –
UnLoc-L [4]	CLIP*	66.10 46.70 – –
CG-DETR [8]	SF+CLIP	67.40 52.10 65.60 45.70
R^2 -Tuning [10]	CLIP	68.71 52.06 – –
VideoLights-B-pt [12]	SF+CLIP+BLIP	72.06 57.94 70.38 51.12
FlashVTG [13]	SF+CLIP	73.10 57.29 72.75 54.33
Mr.BLIP [9]	BLIP-2	76.13 63.35 69.39 55.78
LLaVA-MR [17]	CLIP+LLM	71.85 <u>78.13</u> <u>64.13</u> <u>69.64</u> <u>56.32</u>
SMART	CLIP+LLM	72.03 78.65 65.03 70.46 56.72
QVHighlights Test Set		
SeViLa [54]	BLIP-2	54.50 36.50 – –
UniVTG [16]	SF+CLIP	58.86 40.86 57.60 35.59
QD-DETR [6]	SF+CLIP	62.40 44.98 62.52 39.88
UVCOM [11]	SF+CLIP	63.55 47.47 63.37 42.67
LMR [14]	SF+CLIP	64.40 47.21 64.65 43.16
CG-DETR [8]	SF+CLIP	65.43 48.38 64.51 42.77
R^2 -Tuning [10]	CLIP	68.03 49.35 69.04 47.56
VideoLights-B-pt [12]	SF+CLIP+BLIP	70.36 55.25 69.53 49.17
FlashVTG [13]	SF+CLIP	70.69 53.96 72.33 53.85
Mr.BLIP [9]	BLIP-2	74.77 60.51 68.12 53.38
LLaVA-MR [17]	CLIP+LLM	<u>76.59</u> <u>61.48</u> <u>69.41</u> <u>54.40</u>
SMART	CLIP+LLM	78.15 63.16 70.76 55.54
Charades-STA Test set		
Moment-DETR [3]	–	53.63 31.37 – –
QD-DETR [6]	SF+CLIP	57.31 32.55 – –
LMR [14]	SF+CLIP	55.91 35.19 – –
UniVTG [16]	SF+CLIP	50.10 58.01 35.65 – –
CG-DETR [8]	SF+CLIP	50.13 58.44 36.34 – –
UVCOM [11]	SF+CLIP	59.25 36.64 – –
R^2 -Tuning [10]	CLIP	59.78 37.02 – –
UnLoc-L [4]	CLIP*	60.80 38.40 – –
VideoLights-B-pt [12]	SF+CLIP+BLIP	52.94 61.96 41.05 – –
UniMD+Sync [7]	–	63.98 44.46 – –
InternVideo2-1B [15]	–	68.36 45.03 – –
EaTR [5]	I3D	68.47 44.92 – –
Mr.BLIP [9]	BLIP-2	58.63 69.31 49.29 – –
InternVideo2-6B [15]	–	70.03 48.95 – –
FlashVTG [13]	SF+CLIP	70.32 49.87 – –
LLaVA-MR [17]	CLIP+LLM	<u>59.78</u> <u>70.65</u> <u>49.58</u> <u>69.96</u> <u>39.66</u>
SMART	CLIP+LLM	61.09 72.26 52.17 69.47 40.43

Competing MLLM-based models output a continuous segment from 50s–98s. However, during 88s–96s the man is neither visible nor audible. By incorporating audio cues, SMART detects the absence of speech in that interval and instead predicts two separate segments of 52s–88s and 90s–98s. This matches the ground truth of 52s–88s and 96s–98s much better.

In Example 4, for the query “*People are protesting in Armenia against the government*”, the ground-truth segment consists of multiple intervals: 28s–38s, 62s–72s, and 140s–150s. While scene changes are clear, baseline MLLM models fail to detect the 62s–72s segment. In this portion, the visuals only depict a generic crowd without explicit references to “*Armenia*”. However, the background audio includes a reporter explicitly mentioning “*Armenia*” as the protest location. By leveraging this audio information, SMART captures the missing context and retrieves the correct moments more accurately. The Appendix provides additional qualitative examples with evaluation and detailed discussions.

To better understand where these gains come from, we

next provide ablation, efficiency, and robustness analyses of the proposed audio integration and shot-aware compression design.

C. Ablation Studies and Analysis

We further analyze SMART from four perspectives: the overall contribution of its main components, the role of audio under different query conditions, the stage-wise behavior of Shot-aware Token Compression (STC), and the resulting efficiency–effectiveness trade-off. We finally report hyperparameter studies on audio compression and STC configuration.

Analysis of Audio Contribution. The benefit of audio varies across query types rather than being uniform across all queries. As shown in the supplementary query-type analysis, the majority of queries in QVHighlights are vision-sufficient, while a smaller subset are audio-helpful or strictly audio-dependent. Consistent with this distribution, SMART shows more noticeable improvements on audio-dependent and audio-helpful queries, whereas the gains on vision-sufficient queries are smaller but remain non-negative. This suggests that audio is not universally required, but provides complementary evidence when the target moment is at least partially grounded in acoustic cues.

We further compare several audio integration strategies in Table V. Among them, *Overall Concatenation* performs best, achieving 77.23 R1@0.5, 64.52 R1@0.7, and 70.02 mAP@0.5. In contrast, *Audio-Visual Fusion* performs worse, likely because the additional fusion parameters are harder to optimize under limited training data. *Interleaved Integration* remains competitive in mAP@0.5 but is slightly weaker in recall, suggesting that enforcing fine-grained temporal correspondence may disrupt useful global audio context. Overall, these results indicate that a compact and globally coherent audio representation is more effective than strategies that impose fine-grained temporal alignment.

Stage-wise Analysis of STC. To better understand the role of each stage in STC, we separately evaluate keyframe selection and intra-shot token filtering in Table III. Keyframe selection alone provides a favorable trade-off: it improves coarse localization metrics such as R1@0.5 and mAP@0.75, although the gains are less consistent on stricter metrics such as R1@0.7 and mAP@0.5, indicating that temporal redundancy reduction alone is insufficient without subsequent semantic filtering. In contrast, token filtering alone performs comparably on R1@0.7 but underperforms the audio-enhanced baseline on the other metrics, suggesting that aggressive token compression may discard useful contextual information along with redundant tokens. When the two stages are combined, SMART achieves the best overall performance, demonstrating that intra-shot token filtering is most effective when applied after shot-aware keyframe selection.

Figure 7 provides a qualitative example of this process. Representative keyframes are preserved as full frames, while non-keyframes are passed to the subsequent token filtering stage. Figure 8 further visualizes the temporal feature variation within non-keyframes: tokens with stronger temporal dynamics are retained, whereas weakly varying regions are more

TABLE II
COMPREHENSIVE ABLATION OF THE TWO MAIN COMPONENTS IN SMART ON QVHIGHLIGHTS. BOTH AUDIO INTEGRATION AND SHOT-AWARE TOKEN COMPRESSION IMPROVE OVER THE BASELINE, AND THEIR COMBINATION YIELDS THE BEST OVERALL PERFORMANCE.

	Audio	Shot-aware Token Compression	R1@0.5	R1@0.7	mAP@0.5	mAP@0.75
(a)			76.52	63.23	68.99	55.25
(b)	✓		77.23 (+0.71)	64.52 (+1.29)	70.02 (+1.03)	56.66 (+1.41)
(c)		✓	77.03 (+0.51)	63.48 (+0.25)	69.75 (+0.76)	56.27 (+1.02)
(d)	✓	✓	78.65 (+2.13)	65.03 (+1.80)	70.46 (+1.47)	56.72 (+1.47)

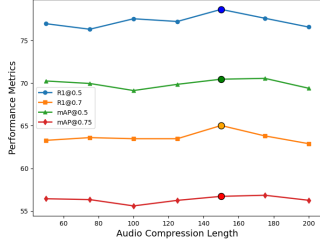


Fig. 5. Impact of **audio compression length** L of the Overall Concatenation strategy on QVHighlights, where the circular marker at $L = 150$ indicates the optimal setting for all metrics.

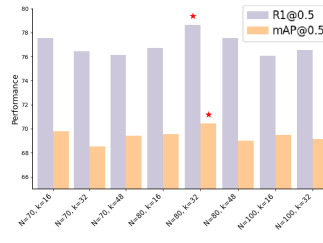


Fig. 6. Determining the optimal **frame number** N and **top- k keyframe** for Shot-aware Token Compression (STC) on QVHighlights, where red stars mark the best configuration ($N = 80$, $k = 32$).

likely to be suppressed. Together, these results support STC as a two-stage compression mechanism rather than a uniform reduction strategy.

Efficiency Analysis. We next examine the computational effect of different STC variants in Table IV. As expected, *Token filtering only* yields the largest raw efficiency gains, reducing the token retention ratio to 56.00%, peak GPU memory to 19.42 GB, and end-to-end latency to 2168.47 ms. However, this aggressive compression also leads to weaker retrieval performance than the full model. SMART is slightly more expensive because selected keyframes are preserved as full frames and only non-keyframes are compressed. Even so, SMART still reduces peak GPU memory from 21.94 GB to 20.72 GB and end-to-end latency from 2451.49 ms to 2224.04 ms relative to the audio-enhanced baseline, while maintaining the strongest retrieval performance. This suggests that SMART provides a more favorable efficiency–effectiveness trade-off than applying token filtering alone.

Hyperparameter Studies. We finally study two implementation choices on QVHighlights. First, Figure 5 analyzes the audio compression length L under the *Overall Concatenation* strategy. We observe that $L = 150$ gives the best overall balance across recall and mAP metrics, suggesting that overly short audio sequences lose useful context, whereas overly long ones introduce redundancy. Second, Figure 6 studies the total frame number N and the number of selected keyframes k for STC. The best configuration is obtained at $N = 80$ and $k = 32$, which balances visual coverage and sequence efficiency. Too few keyframes lead to excessive compression, whereas too many reintroduce redundancy without consistent gains.

V. CONCLUSION

We introduce **Shot-aware Multimodal Audio-enhanced Retrieval of Temporal Segments (SMART)**, a multimodal moment retrieval framework that improves temporal localization by incorporating complementary audio cues together with a **Shot-aware Token Compression** strategy. SMART first performs shot-aware keyframe selection to preserve representative frames as full visual observations, and then applies token filtering only to non-keyframes to reduce redundant visual content while retaining temporally informative regions. This two-stage design balances efficiency and effectiveness, enabling more efficient long-sequence processing while preserving retrieval performance. In parallel, compact audio representations provide complementary semantic cues that are particularly helpful when the target moment is at least partially grounded in acoustic evidence. Extensive experiments on Charades-STA and QVHighlights show that SMART consistently improves over strong baselines while achieving favorable performance–efficiency trade-offs.

Limitations: Despite its strong performance, SMART still has several limitations. First, the current audio branch relies on compressed global audio context rather than explicitly modeling fine-grained audio–visual temporal alignment, which may limit its ability to capture subtle audio-event timing in complex scenes. Second, although the proposed shot-aware compression preserves selected keyframes and only compresses non-keyframes, subtle visual details may still be weakened under aggressive compression or highly dynamic motion patterns. Third, while our additional analyses show that SMART is reasonably robust to moderate shot-boundary perturbations and can transfer its visual compression design to another backbone, its behavior under larger distribution shifts or broader open-world settings remains an important direction for future work.

Future Work: SMART offers several opportunities for improvement. Dense audio-text alignment could enhance grounding in complex scenes, while extending SMART to open-world or zero-shot retrieval could improve generalization to unseen events and domains. Integrating additional modalities or hierarchical temporal modeling may further strengthen robust, scalable multimodal moment retrieval.

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TABLE III

STAGE-WISE ABLATION OF STC ON QVHighlights. KEYFRAME SELECTION CONSISTENTLY HELPS, WHILE TOKEN FILTERING ALONE IS LESS STABLE. THEIR COMBINATION IN SMART ACHIEVES THE BEST OVERALL PERFORMANCE.

Variant	Keyframe Selection	Intra-shot Token Filtering	R1@0.5	R1@0.7	mAP@0.5	mAP@0.75
Keyframe only	✓		77.35	63.81	69.95	56.76
Token filtering only		✓	76.90	64.19	69.21	55.94
SMART	✓	✓	78.65	65.03	70.46	56.72

TABLE IV

EFFICIENCY COMPARISON ON QVHighlights.

TOKEN FILTERING ONLY ACHIEVES THE LARGEST EFFICIENCY GAINS, WHILE SMART OFFERS A STRONGER EFFICIENCY-EFFECTIVENESS TRADE-OFF.

Method	Token Kept %	Peak GPU Mem GB	E2E Latency ms	Throughput samples/s
Baseline with audio	100	21.94	2451.49	0.4331
Keyframe only	88.75	21.29	2368.54	0.4456
Token filtering only	56.00	19.42	2168.47	0.4924
SMART	78.89	20.72	2224.04	0.4750

TABLE V

ABLATION STUDY OF DIFFERENT AUDIO CONCATENATION STRATEGIES ON THE QVHighlights DATASET.

Audio	R1@0.5	R1@0.7	mAP@0.5	mAP@0.75
Overall Concatenation	77.23	64.52	70.02	56.66
Audio-Visual Fusion	24.84	16.23	20.47	12.26
Interleaved Alignment	76.71	63.16	70.01	56.12
Voice + Ambient Sound	76.00	63.61	69.82	56.83

A family is playing basketball together on a green court outside.

GT
SMART

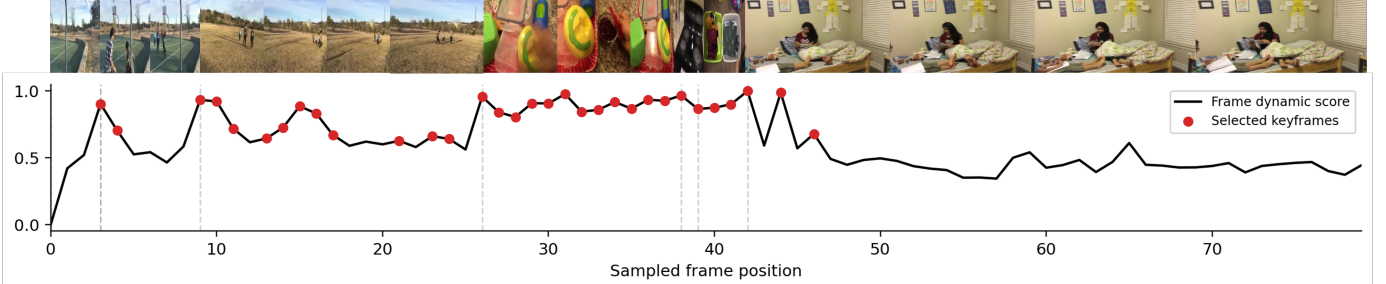


Fig. 7. **Example of shot-aware keyframe selection on QVHighlights [3].** Our keyframe selection stage preserves representative keyframes from each shot as full frames, while the remaining non-keyframes are processed by the subsequent token filtering stage. This design retains the main temporal evidence while avoiding unnecessary redundancy.



Fig. 8. **Temporal feature variation heatmaps for two adjacent non-keyframes.** The heatmaps are computed from LayerNorm-normalized visual encoder patch features before STC compression. For each sampled frame t and spatial patch p , we compute $H_t(p) = \|z_{t,p} - z_{t-1,p}\|_2$, where $z_{t,p}$ denotes the visual encoder feature of patch p at frame t . The CLS token, if present, is excluded before reshaping the scores into a spatial grid. Warmer colors indicate stronger adjacent-frame feature changes. These maps are not attention maps; they visualize local temporal dynamics. The actual intra-shot token filtering operates on Q-Former visual query tokens after T5 projection, using $s(q) = \frac{1}{D} \sum_{d=1}^D \text{Std}_{t \in S} (h_{t,q,d})$ within each shot S , and retains tokens with the highest $s(q)$ values.

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